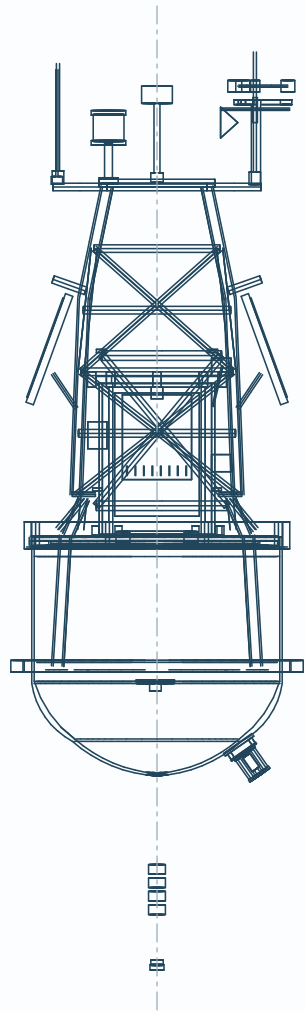


## PROJECT FALCON-01 — COASTAL MONITORING BUOY

MASTER ASSEMBLY · SUBSYSTEMS · ELECTRONICS · POWER · MOORING · V2 CAD REFERENCE

**REFERENCE / NOT FOR FABRICATION**

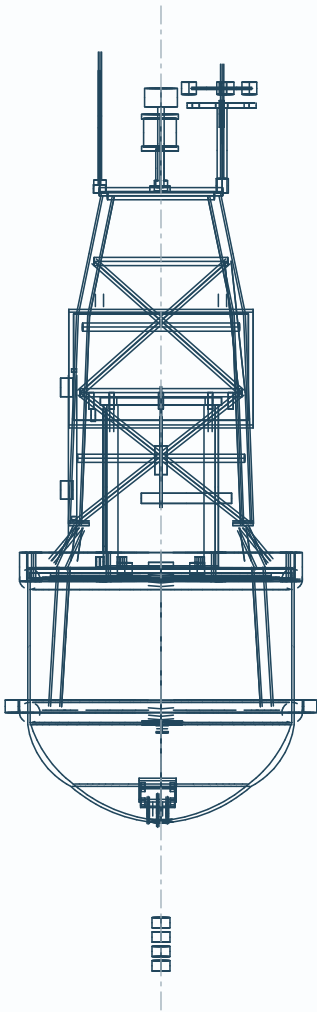
### A — FRONT ELEVATION · CAD-PROJECTED



FLOAT Ø0.650 m · V2 CAD REF

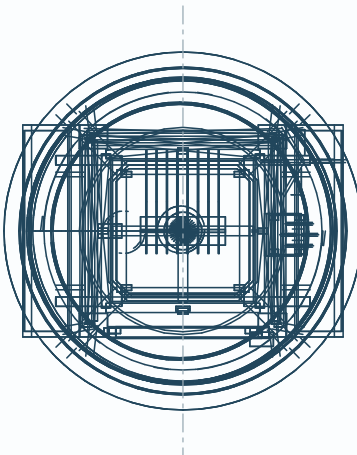
Overall installed height and loaded waterline: VERIFY IN NATIVE CAD / FLOTATION TEST

### B — RIGHT ELEVATION

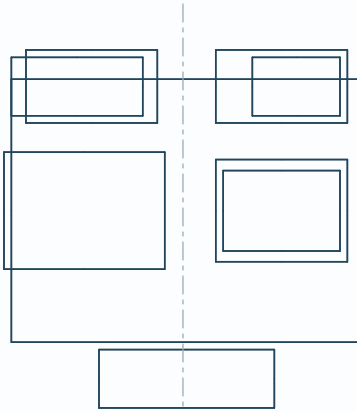


ORTHOGRAPHIC · DO NOT SCALE

### C — PLAN VIEW

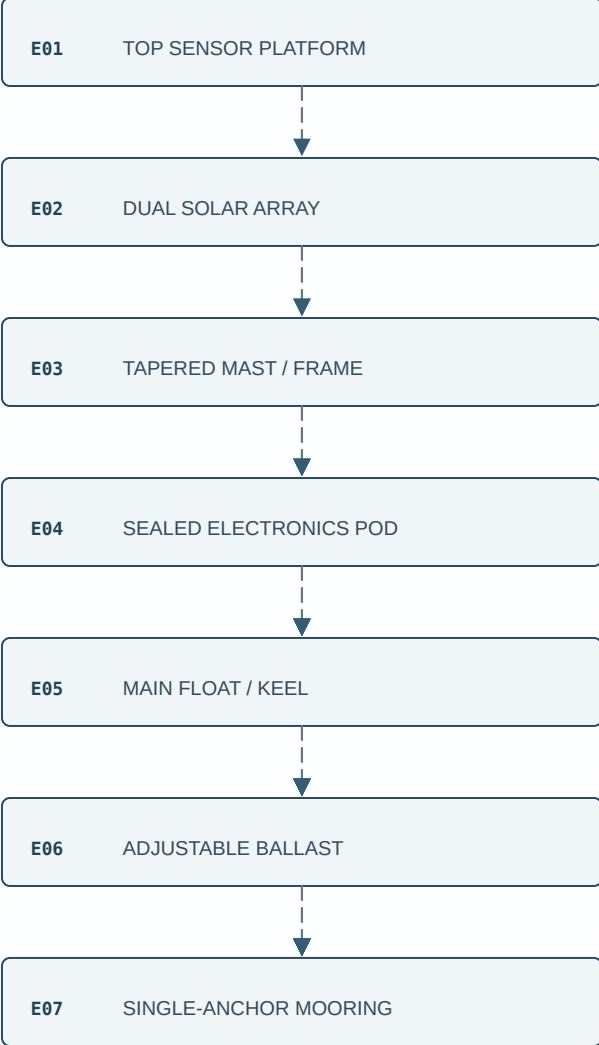


## D — ELECTRONICS POD / EQUIPMENT PLAN



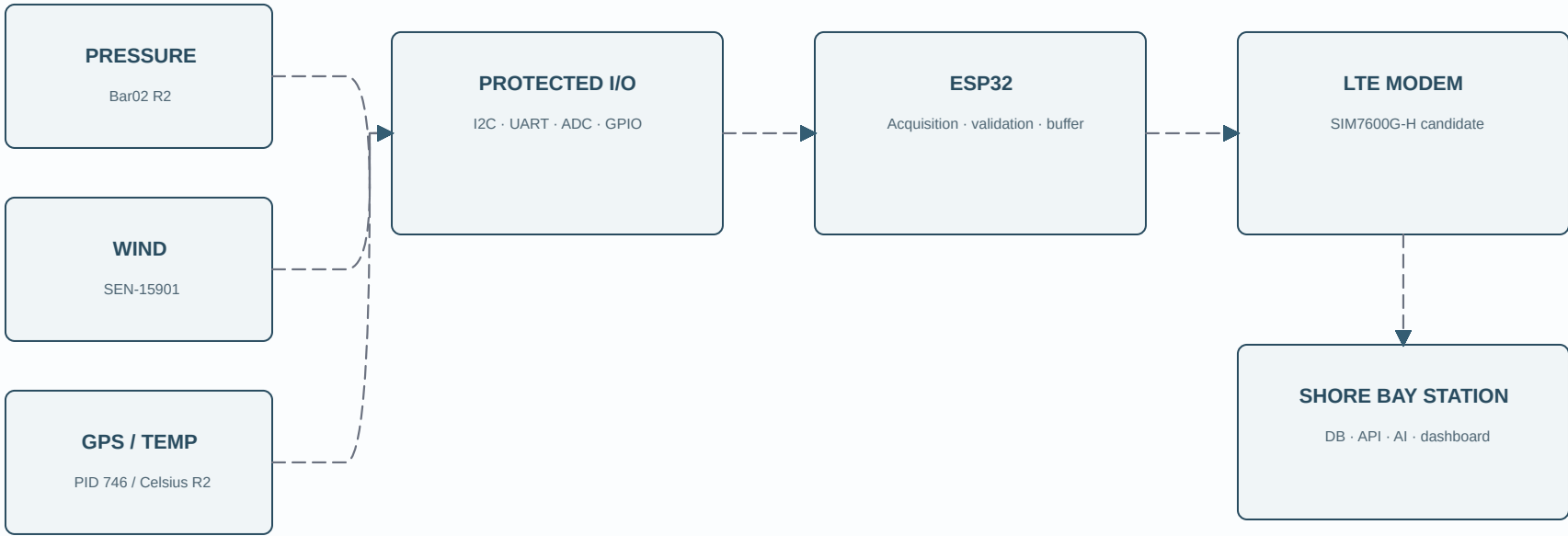
Orange Pi excluded - shore Bay Station only

## E — ASSEMBLY HIERARCHY · NTS



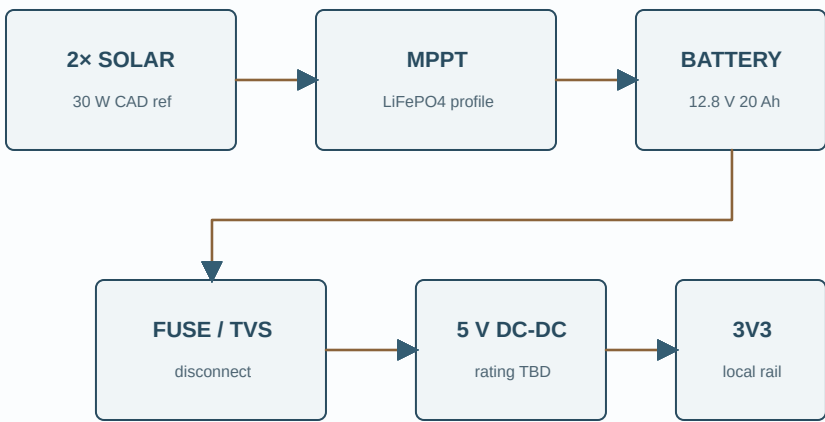
Hierarchy only; separation distances are not dimensions

## F — SUBSYSTEM INTERCONNECTION / DATA FLOW



Pressure-derived estimated wave height · AI prediction is processed at the shore Bay Station.

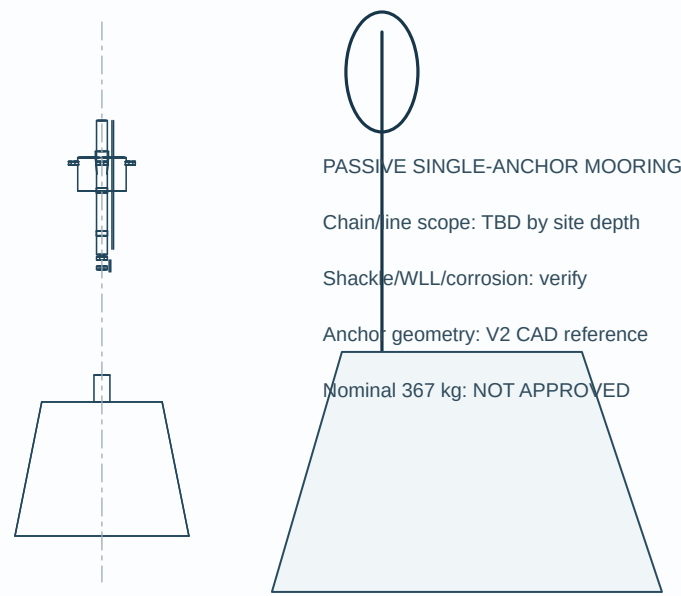
## G — POWER MANAGEMENT



INA260 battery 0x40 · INA260 solar 0x41

### Final fuses, converters and autonomy require measured load tests

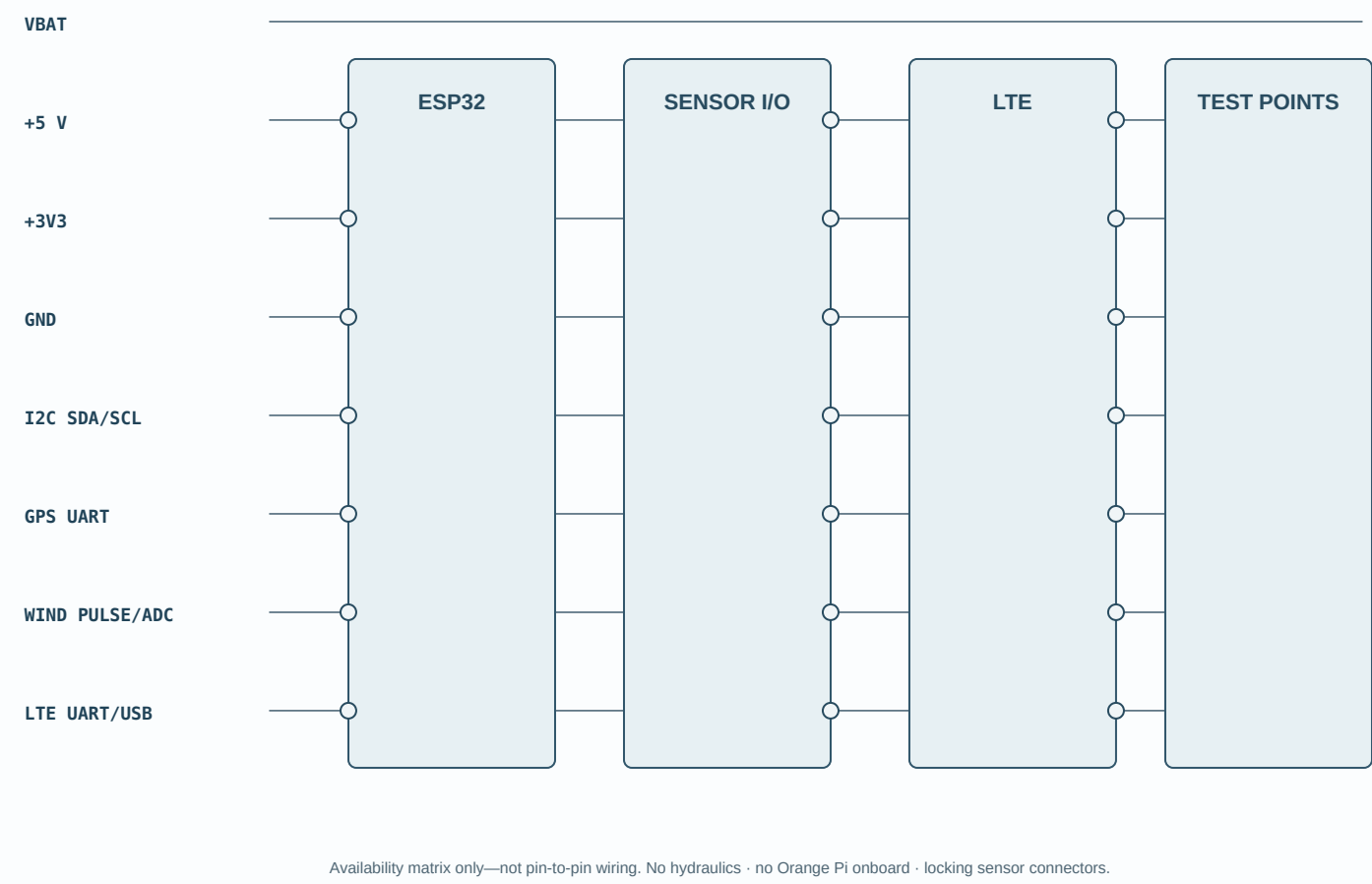
## H — MOORING / ANCHORAGE



## J — CONTROLLED COMPONENT REGISTER

| ID  | COMPONENT                     | STATUS     | FUNCTION               |
|-----|-------------------------------|------------|------------------------|
| S01 | Blue Robotics Bar02 R2        | PROTOTYPE  | Pressure / wave input  |
| S02 | Adafruit Ultimate GPS PID 746 | PROTOTYPE  | Position / geofence    |
| S03 | SparkFun SEN-15901            | PROTOTYPE  | Wind speed / direction |
| S04 | Blue Robotics Celsius R2      | CANDIDATE  | Water temperature      |
| H01 | Adafruit MCP9808 PID 1782     | SELECTED   | Enclosure temperature  |
| H02 | Adafruit INA260 PID 4226 ×2   | PROTOTYPE  | Battery / solar health |
| C01 | ESP32-DevKitC V4              | SELECTED   | Buoy controller        |
| C02 | SIM7600G-H 4G HAT             | CANDIDATE  | Cellular telemetry     |
| P01 | LiFePO4 12.8 V 20 Ah          | RATING SET | Exact SKU / case TBD   |
| SEC | Contact switch PID 375        | PROTOTYPE  | Enclosure-open state   |
| A01 | Buzzer + driver               | TBD        | Audible security alarm |

## K — CONTROLLED INTERFACE MATRIX · NOT A NETLIST



Availability matrix only—not pin-to-pin wiring. No hydraulics · no Orange Pi onboard · locking sensor connectors

## L — GENERAL NOTES

1. Orthographic linework is projected from PROJECT-FALCON-V2.glb.
2. V2 geometry remains a proposed replacement reference.
3. Do not scale this drawing; verify native CAD and purchased parts.
4. Dimensions, mass and placement marked TBD must not be invented.
5. Complete stability, structure, ingress, thermal and mooring reviews.
6. Orange Pi/Bay Station computer remains ashore.

## PRELIMINARY ENGINEERING REFERENCE

## M — DRAWING CONTROL

|                                  |                                      |
|----------------------------------|--------------------------------------|
| PROJECT                          | DRAWING TITLE                        |
| <b>FALCON-01</b>                 | <b>MASTER ASSEMBLY &amp; SYSTEMS</b> |
| DRAWING NO. <b>FALCON-BP-000</b> | REVISION <b>P0</b>                   |
| SCALE <b>NTS</b>                 | DATE <b>2026-09-01</b>               |
| SOURCE <b>V2 GLB + DOCS</b>      | STATUS <b>REFERENCE</b>              |

## RELEASE GATES / APPROVAL

- ☐ Native CAD dimension audit ☐ Purchased-part fit / connector clearance ☐ Loaded mass / displacement / waterline / freeboard
- ☐ CG / stability / righting test ☐ Structural load and fastener review ☐ IP / leak / thermal / corrosion validation
- ☐ Solar / battery measured load budget ☐ Pressure installation and wave-reference validation ☐ Mooring WLL / anchor design review

PREPARED BY / DATE

ENGINEERING REVIEW / DATE

THESIS ADVISER APPROVAL / DATE

**NOT FOR FABRICATION**